MIL-C-47138(MI)
24 May 1974
SUPERSEDING
MIS-15729
29 December 1965

MILITARY SPECIFICATION

COATING, POLYVINYL CHLORIDE

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a tough durable polyvinyl chloride coating which is used for preventing shedding of loose fibers from the surface and edges of insulation blankets. It is also suitable for corrosion and abrasion resistance on various surfaces such as steel, wood, and concrete.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

TT-P-143	Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing and Marking of

STANDARDS

Federal

FED-STD-141	Paint, Varnish, Lacquer and Related Materials, Methods of Inspection, Sampling and Testing

FSC 9330

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Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

Uniform Classification Committee

Uniform Freight Classification Rules

(Application for copies should be addressed to Uniform Classification Committee, 202 Chicago Union Station, Chicago, Illinois 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Preproduction sample. When specified by the procuring activity (see 6.2), a preproduction sample shall be prepared using the same methods proposed for subsequent production and shall be subjected to preproduction inspection (see 4.3).

3.2 Material. The material is a ketone solution of polyvinyl chloride with a plasticizer additive.

3.3 Properties.

3.3.1 Percent solids. The percent solids shall be 13 plus or minus .5 percent.

3.3.2 Weight per gallon. The weight per gallon shall be 8.82 plus or minus .02 pounds.

3.3.3 Viscosity. The viscosity shall be 75 plus or minus 15 seconds.

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3.3.4 Heat resistance. The material shall be capable of withstanding a dry temperature of 250 plus or minus 5 degrees Fahrenheit (F) or a wet immersion of 155 plus or minus 5 degrees F for seven days without evidence of sag, wrinkles, or other detrimental effects.

3.4 Color. Unless otherwise specified the color shall be gray. Other colors available are white, black, and clear.

3.5 Workmanship. The workmanship shall be such as to insure a product which is uniform and in conformance with this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification and tests. The testing of the material shall be classified as follows:

4.2.1 Preproduction inspection (4.3).

4.2.2 Quality conformance inspection (4.4).

4.3 Preproduction inspection. Preproduction inspection shall consist of all examinations and tests specified herein and shall be conducted only on preproduction samples. The preproduction samples shall be prepared using the same methods to be used for the manufacture of all subsequent production material.

4.4 Quality conformance inspection.

4.4.1 Tests. Quality conformance inspection tests shall be the examination and tests required for each production lot. A lot shall be defined as that quantity of material which has been manufactured at one time by a chemical or physical process.

4.4.2 Sampling. A random sample of filled containers of polyvinyl chloride shall be selected from each lot in accordance with Military Standard MIL-STD-105 at Inspection Level 1 and an acceptable quality

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level (AQL) of 2.5 percent defective to verify compliance with Section 3.

4.4.3 Inspection. Quality conformance inspection shall include the tests and requirements specified per 3.3.1, 3.3.3, and 3.5. Failure of any of the tests shall be cause for rejection of the production lot.

4.5 Test methods.

4.5.1 Examination of product. The coating shall be examined for conformance with the material, color, and workmanship requirements of this specification. (See 3.2, 3.4, and 3.5.)

4.5.2 Percent solids. The percent solids shall be determined by testing in accordance with Method 4041 of FED-STD-141. (See 3.3.1.)

4.5.3 Weight per gallon. Weight per gallon shall be determined by testing in accordance with Method 4184 of FED-STD-141. (See 3.3.2.)

4.5.4 Viscosity. Viscosity shall be determined in accordance with Method 4282 of FED-STD-141 using a number 4 Ford cup. (See 3.3.3.)

4.5.5 Heat resistance.

4.5.5.1 Wet immersion. Resistance to wet immersion shall be determined by immersing the cured film in water held at 155 plus or minus 5 degrees F for seven days. Visual inspection shall determine conformity to 3.3.4.

4.5.5.2 Dry exposure. Dry exposure resistance shall be accomplished by subjecting the cured film to 250 plus or minus 10 degrees F for seven days. Visual inspection shall then determine conformity to 3.3.4.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.2) in accordance with TT-P-143.

5.2 Packing. Packing shall be Level A, B, or C as specified (see 6.2) in accordance with TT-P-143.

5.3 Marking. Marking shall be in accordance with MIL-STD-129 and shall include but shall not necessarily be limited to the following:

a. This specification number.

b. Vendor's name.

c. Manufacturer's material designation.

d. Date of manufacture.

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e. Quantity.

f. Contract or procurement document.

g. Shelf life.

6. NOTES

6.1 Intended use. This surface coating is to be used as a coating on insulation blankets to prevent shedding of loose fibers and as abrasion resistance for surfaces of metal.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether preproduction samples and test reports are required.

c. Special packaging and packing instructions, if required.

d. Other special instructions.

e. Selection of applicable levels of packaging and packing.

6.3 Supersession data. This specification includes the requirements of Missile Interim Specification MIS-15729, dated 29 December 1965.

Custodian:
ARMY-MI

Preparing Activity:
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Project No. 9330-A625

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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